INCH-POUND
MIL-W-48855C (AR)
AMENDMENT 4
28 FEBRUARY 1989
SUPERSEDING
AMENDMENT 3
17 December 1987

MILITARY SPECIFICATION

WARHEAD BODY ASSEMBLY
FOR
WARHEAD SECTION, GM, HE, M258

This amendment forms a part of Military Specification MIL-W-48855C (AR), dated 17 May 1985, and is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1, STANDARDS: Add the following:

"FEDERAL

FED. TEST METHOD STD. No. 151 -Metals; Test Methods"

2.1.2, Drawings: Add the following:

"9297072 -Preform, Powder Metal "

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3.4, Add the following:

"Isolation with warhead case and P3-9/P3-2 shorted ≥1 meg ohm".

* 3.5: Delete in its entirety and substitute the following:

"3.5 Process control document. The contractor shall supply a process control document which is to include descriptions of all production processes, procedures and equipment (including in-process inspection equipment) used to manufacture and inspect the production material. These descriptions shall identify the basic fabrication process, fabrication equipment (including methods for evaluating status of equipment), controls associated

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fabrication processes and in-process inspections. All initial and revised process control documents shall be submitted to the technical office for review and approval of completeness prior to its use in production (see 6.4). It should be noted that it is not the government's intention to approve the contractor's process itself, but merely to review the completeness and adequacy of the process control document."

The contractor shall make available, at the time of government inspection, all data necessary to show compliance to the process control document (ie. Temperatures, Pressures, Times, Speeds, Weights, etc.)."

3.7, line 5: Add the following after 'cracks', ", porosity".

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4.4.1, Sentence 4: Delete in its entirety and substitute:

"Additionally, preform lot size shall be limited to the quantity sintered in a single oven at one time, and for a batch process, shall be limited to a single batch."

4.4.2, b.: Delete column labeled 'DENSITY**' and substitute:

"DENSITY**
10% of lot/
minimum
10% of lot
10% of lot
10% of lot"

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4.4.2, c: Delete column labeled 'DENSITY**' and substitute:

"DENSITY**
5% of lot/
minimum
5% of lot

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4.4.2.2: Add new major categories in the appropriate columns as follows:

"Category Major	Exam or Test	AQL or 100%	Req. Para.	Para Ref/ Insp. Meth.
109	Presence of threads in connector center pin.	100%	3.1	Gage
110	Torque test of electrical plug connector.	100%	3.1	Gage

"

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4.4.2.4: Add new categories in the appropriate columns as follows:

"Category Major	Exam or Test	AQL or 100%	Req. Para.	Para REf/ Insp. Meth.
105	Chemical analysis (PMP)	2/	3.1	4.5.10"

4.4.2.4: Add new note as follows:

"2/ A chemical analysis shall be performed on each lot."

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* 4.4.2.5, Major 106: Delete "FWD and AFT threads" and substitute " AFT threads and AFT face".

* 4.4.2.5, Major 111: Delete "FWD and AFT threads" and substitute "AFT threads and AFT face".

* 4.4.2.5, Major 112: Delete "FWD and AFT threads" and substitute "AFT threads and AFT face".

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* 4.4.2.5, Major 137: Delete "-" and substitute "+".

* 4.4.2.5, Major 141: Delete ".010 and substitute ".002".

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4.4.2.5, Minor 208: Delete "±" and substitute"+".

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4.5.1.2: Delete in its entirety and substitute the following:

"4.5.1.2 Powder metal preform. The test specimens for a warhead lot that are to be machined from powder metal preforms shall be obtained from one of the following methods.

a. All specimens from the first method are obtained from the sidewall of the powder metal preform. The test specimens shall be produced to such dimensions so that yield and impact test bars shall be machined to the form and dimensions shown in figures 4 and 5. Two test specimens of each type shall be machined from each body.

b. The yield specimens shall be obtained from the sidewall while the impact specimens shall be obtained from the base of the powder metal preform. The test specimens shall be produced of such dimensions so that the yield and impact test bars shall be machined to the form and dimensions shown in figures 4 and 5. Two test specimens of each type shall be machined from each body."

4.5.2: Delete in its entirety and substitute the following:

"4.5.2 Impact resistance of warhead body."

4.5.2.1 Forged Warhead body. Charpy v-notch impact test shall be conducted in accordance with ASTM E23. The temperature of the test specimen at the time of the fracture shall be -40 ±2°F. The error in the Charpy machine shall not exceed ±1 ft-lb, when the machine is tested using comparison specimens prepared by US Army Materials Research Agency (see 6.6). This comparison shall have been made within one year prior the time of inspection testing.

4.5.2.2 Powder metal preform. The temperature of the test specimen at the time of the fracture shall be -40 ±2°F. The error in the Charpy machine shall not exceed ±1 ft-lb, when the machine is tested using comparison specimens prepared by US Army Materials Research Agency (see 6.6). This comparison shall have been made within one year prior to the time of the inspection testing.

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4.5.3.2, Sentence 1: Delete in its entirety and substitute:

"Tension tests shall be accomplished in accordance with ASTM E8, except that the configuration shall be as per figure 5, and the rate of strain will be 0.003 to 0.007 inch per minute through the yield strength and then to be increased so as to produce failure in approximately one additional minute."

4.5.8: Add the following:

"The HTS circuit isolation shall be tested at 200 VDC with a pressure of 10 PSI applied to the HTS shield."

4.5.10: Add new paragraph 4.5.10 as follows:

"4.5.10 Chemical analysis (Powder metal only). A chemical analysis shall be performed on each lot of preforms produced and should identify all elements shown to be crucial by the drawing. The chemical analysis shall be performed in accordance with Method 111 or Method 112 of FED-STD-151, or another method approved by the procuring activity."

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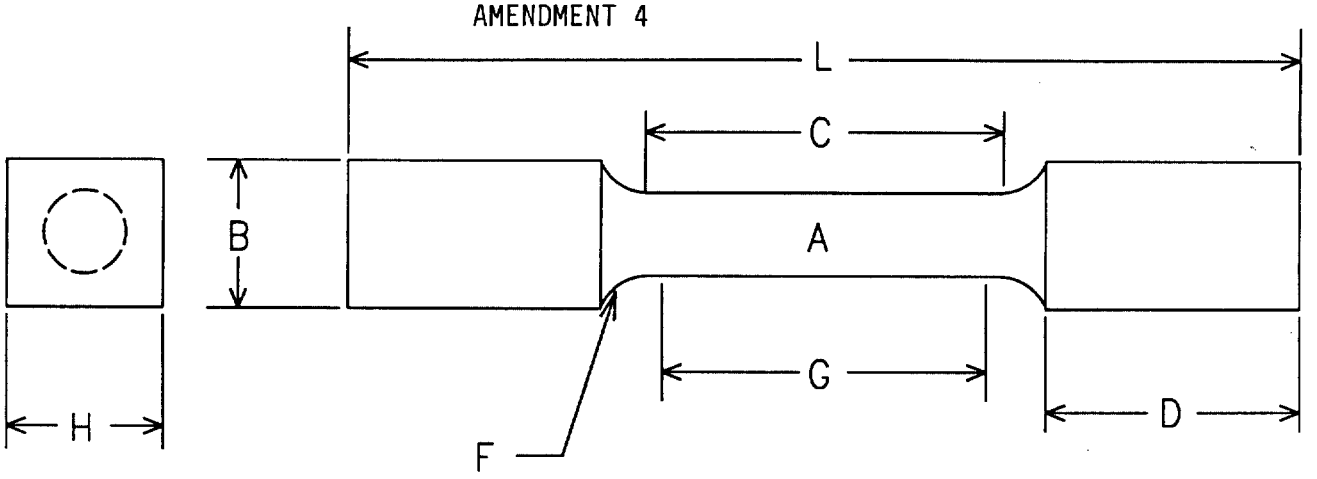
Delete "Figure 1. Location of test specimens in warhead body" and substitute "Figure 1. Location of test specimens in forged warhead body".

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Add new Page 23A.

Add new Page 23B.

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ALL DIMENSIONS IN INCHES

DIMENSIONS	
A- DIAMETER AT CENTER[1]	0.157 ± 0.001
B- WIDTH OF GRIP ENDS APPROX.[3]	0.250
C- LENGTH OF REDUCED SECTION MIN.	1.00
D- GRIP LENGTH, APPROX.[3]	1,250
F- FILLET RADIUS, MIN.[4]	0.250
G- GAGE LENGTH	0.750 ± 0.005
H- DEPTH OF GRIP ENDS APPROX.	0.250
L- TOTAL LENGTH, APPROX.[2,3]	4.00
AREA OF CROSS SECTION SQUARE INCHES APPROX.	1/2

1. Diameter of the reduced section may be smaller at the center than at ends. Difference shall not exceed 1 percent of the diameter at the center.

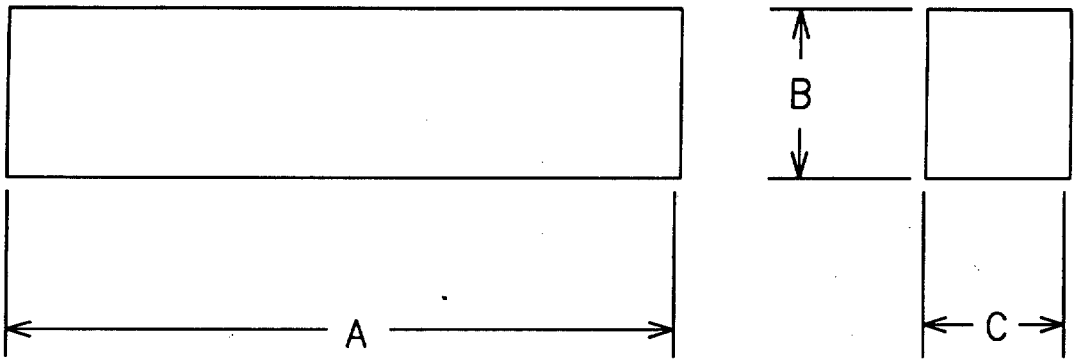
2. When extensometer is required to determine elastic properties, dimensions C and L may be modified to provide therefor. In all cases the percentage of elongation shall be based on dimension G.

3. Ends may be of any shape to fit the holders of the testing machine such that the preform is tested axially.

4. A generous fillet is desirable.

FIGURE 4. Powder metal preform tensile specimen

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DIMENSION	MILLIMETERS	TOLERANCE + MILLIMETERS -	INCHES	TOLERANCE + INCHES -
A- LENGTH	55.000	0.0 2.50	2.165	0.0 0.100
B- WIDTH	3.912	0.025 0.025	0.154	0.001 0.001
C- DEPTH	3.912	0.025 0.025	0.154	0.001 0.001

FIGURE 5. Powder metal preform impact specimen

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6.3, Line 2: Delete ",Dover, NJ 07801." and substitute the following "(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirement List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply."

6.4: Delete in its entirety and substitute the following:

"6.4 Process control document. The process control document shall be submitted in accordance with the process control document requirements of data item description DI-R-5297A (Tailored), to Commander, U. S. Army Armament Munitions and Chemical Command, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000 prior to the start of production. Questions, if any, regarding proprietary issues of the process control documentation may be directed to the command legal office ATTN: AMSMC-GC at the above address."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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